Section 14.8: Lagrange Multipliers

Partial Derivatives

Why Constrained Optimization?

In Section 14.7 we found absolute extrema by checking critical points **and** the boundary.

But what if the constraint is more complicated than a rectangular box?

Real-world examples:

- Maximize profit subject to a fixed budget
- Minimize material used to build a container with a required volume
- Find the hottest point on a wire bent into a curve

In each case we want to optimize $f(x, y)$ (or $f(x, y, z)$) but we are **restricted** to points satisfying a constraint $g(x, y) = k$.

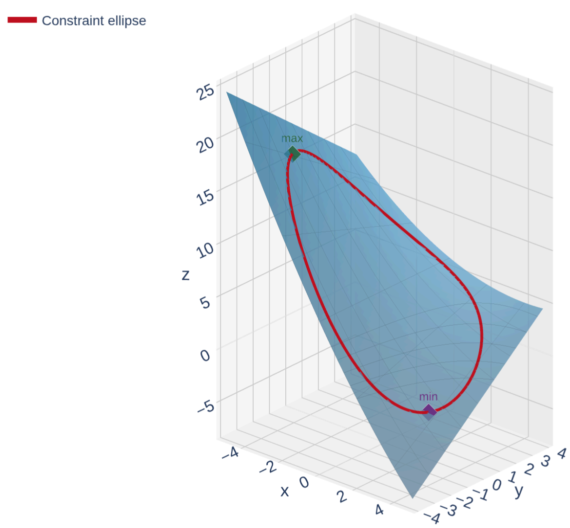
> Lagrange multipliers give us an elegant, systematic method for optimizing a function **subject to a constraint** — no parameterization needed.

A Motivating Example

Example 1

Find the absolute maximum and minimum of $f(x, y) = 6 - 2x + 0.1x^2 + 0.3xy$ on the ellipse $0.2x^2 + 0.3y^2 = 3.6$.

The **constraint** is that we only consider points (x, y) lying on the ellipse. We can't just find critical points of f — the extrema occur *on the ellipse*.



The surface $z = f(x, y)$ restricted to the ellipse | Interactive version

Let's look at two approaches to this problem.

Approach 1: Parameterize the Constraint

Idea: Turn the constraint curve into a parametric curve, then substitute into f to get a single-variable problem.

The ellipse $0.2x^2 + 0.3y^2 = 3.6$ can be parameterized as:

$$x = \sqrt{18}\cos t, \quad y = \sqrt{12}\sin t, \quad 0 \le t < 2\pi$$

Now substitute into f:

$$f(t) = 6 - 2\sqrt{18}\cos t + 1.8\cos^2 t + 0.3\sqrt{216}\cos t \sin t$$

This is a **single-variable** calculus problem — find the absolute max/min on $[0, 2\pi)$. But the algebra is *ugly*.

Parameterization **works** but has drawbacks: finding a parameterization can be difficult, and the resulting function can be messy. We need a better way.

The Geometry Behind Lagrange Multipliers

Think about walking along the constraint curve $g(x, y) = k$. At most points, f is either increasing or decreasing as you walk.

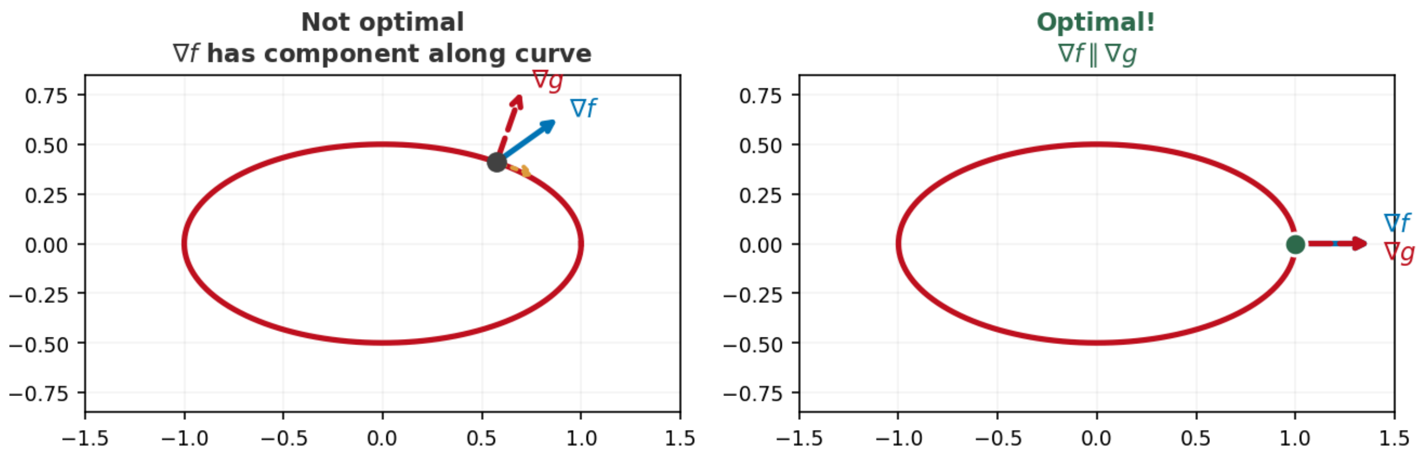
At an **extremum** along the constraint, f is neither increasing nor decreasing — the level curves of f are **tangent** to the constraint curve.

Recall:

- ∇f is **perpendicular** to level curves of f
- ∇g is **perpendicular** to the constraint curve $g = k$

If both gradients are perpendicular to the same curve at the same point, they must be **parallel**:

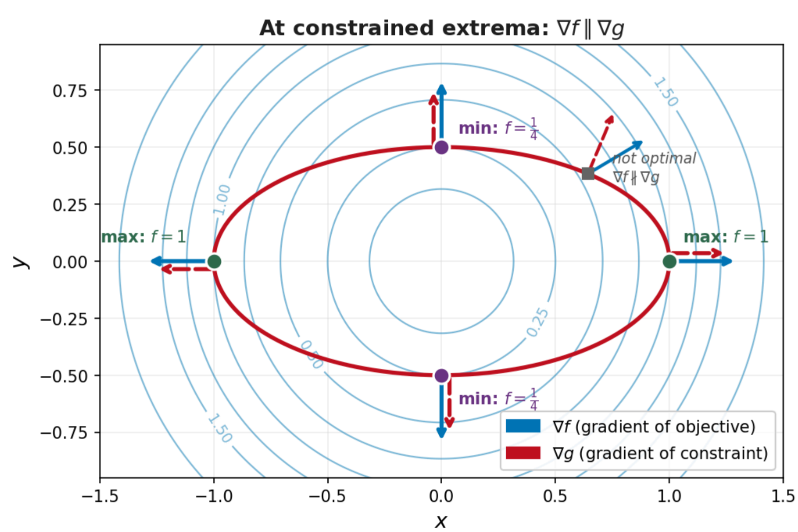
$$\nabla f = \lambda \nabla g$$



At a constrained extremum, ∇f and ∇g point in the **same (or opposite) direction**. The scalar λ is called the **Lagrange multiplier**.

See It on a Contour Map

Here's the geometry in action: level curves of $f(x, y) = x^2 + y^2$ (blue circles) with the constraint ellipse $x^2 + 4y^2 = 1$ (red).



At the **green dots** (max), the level curve $f = 1$ is tangent to the ellipse, and $\nabla f \parallel \nabla g$.

At the **purple dots** (min), the level curve $f = \frac{1}{4}$ is tangent to the ellipse, and $\nabla f \parallel \nabla g$.

At the **gray square** (not optimal), ∇f and ∇g point in different directions — you can still increase f by walking along the ellipse.

The Method of Lagrange Multipliers

Method of Lagrange Multipliers

To find the maximum and minimum values of $f(x, y)$ subject to the constraint $g(x, y) = k$ (assuming these extreme values exist and $\nabla g \neq \vec{0}$ on the curve $g = k$):

1. Find all values of x, y, and λ such that

$$\nabla f(x, y) = \lambda \nabla g(x, y) \quad \text{and} \quad g(x, y) = k$$

1. Evaluate f at all the points (x, y) found in step 1. The largest value is the **maximum**; the smallest is the **minimum**.

In two variables, $\nabla f = \lambda \nabla g$ becomes the system:

$$f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad g(x, y) = k$$

That's **3 equations in 3 unknowns** (x, y, λ).

You often don't need the actual value of λ — it can be eliminated from the system to find x and y.

Quick Check: Set Up a Lagrange System

Suppose you want to minimize $f(x, y) = x^2 + 2y^2$ subject to $x + y = 4$.

Write the three equations you'd need to solve (but don't solve them).

Identify f and g, compute ∇f and ∇g, then write $f_x = \lambda g_x$, $f_y = \lambda g_y$, and $g = k$.

$f(x, y) = x^2 + 2y^2, \quad g(x, y) = x + y = 4.$
$\nabla f = \langle 2x,\ 4y \rangle, \quad \nabla g = \langle 1,\ 1 \rangle.$

The system:

$$2x = \lambda(1), \quad 4y = \lambda(1), \quad x + y = 4$$

Setting up the system is the mechanical part. The thinking comes in solving it — look for ways to eliminate λ.

Example 2: Closest and Farthest from the Origin

Example 2

Find the maximum and minimum values of $f(x, y) = x^2 + y^2$ on the ellipse $x^2 + 4y^2 = 1$.

Since $x^2 + y^2$ is the square of the distance from the origin, maximizing/minimizing f finds the **farthest** and **closest** points on the ellipse to the origin.

Set up: $f(x, y) = x^2 + y^2, \quad g(x, y) = x^2 + 4y^2 = 1.$

Write $\nabla f = \lambda \nabla g$ and $g = 1$. You'll get a system with cases — set factors to zero.

Compute gradients: $\nabla f = \langle 2x,\, 2y \rangle, \quad \nabla g = \langle 2x,\, 8y \rangle.$

System: $2x = \lambda(2x), \quad 2y = \lambda(8y), \quad x^2 + 4y^2 = 1.$

From equation 1: $2x(1 - \lambda) = 0 \Rightarrow x = 0$ or $\lambda = 1$.

Case 1: $\lambda = 1$**.** Equation 2 gives $2y = 8y \Rightarrow y = 0$.

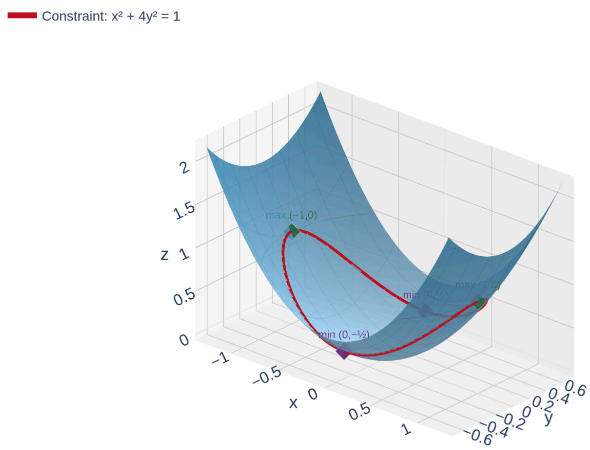
Constraint: $x^2 = 1 \Rightarrow x = \pm 1$. Points: $(\pm 1, 0)$.

Case 2: $x = 0$**.** Constraint: $4y^2 = 1 \Rightarrow y = \pm \frac{1}{2}$. Points: $(0, \pm \frac{1}{2})$.

Evaluate:

- $f(\pm 1, 0) = 1 \leftarrow$ **maximum** (farthest from origin)
- $f(0, \pm \frac{1}{2}) = \frac{1}{4} \leftarrow$ **minimum** (closest to origin)

The case analysis strategy is typical: factor one equation, then consider each factor being zero. This gives distinct cases to chase through the system.



Interactive version

The farthest points $(\pm 1, 0)$ sit at the ends of the ellipse; the closest points $(0, \pm \frac{1}{2})$ sit at the narrow waist — exactly what the contour map from Slide 5 predicted.

Your Turn: Lagrange Practice

Example 3

Find the maximum and minimum values of $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$.

Hint: Set up $\nabla f = \lambda \nabla g$ where $g(x, y) = x^2 + y^2$.

Set up the three equations, then solve for x, y, and λ.

Gradients: $\nabla f = \langle 3,\, 4 \rangle, \quad \nabla g = \langle 2x,\, 2y \rangle$.

System: $3 = 2\lambda x, \quad 4 = 2\lambda y, \quad x^2 + y^2 = 1$.

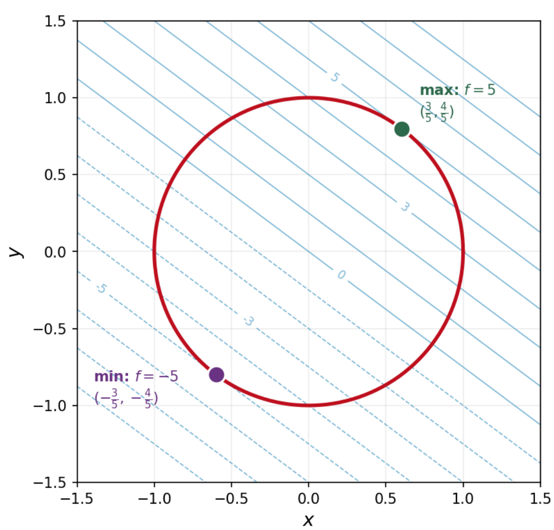
Solve for x and y: $x = \dfrac{3}{2\lambda}, \quad y = \dfrac{2}{\lambda}$.

Substitute into constraint:

$$\frac{9}{4\lambda^2} + \frac{4}{\lambda^2} = 1 \quad \Rightarrow \quad \frac{25}{4\lambda^2} = 1 \quad \Rightarrow \quad \lambda = \pm\frac{5}{2}$$

$\lambda = \frac{5}{2}$: $x = \frac{3}{5}$, $y = \frac{4}{5}$. $f = 5$. $\leftarrow$ **maximum**

$\lambda = -\frac{5}{2}$: $x = -\frac{3}{5}$, $y = -\frac{4}{5}$. $f = -5$. $\leftarrow$ **minimum**



Notice how the level lines of f (parallel lines) are **tangent** to the circle at the optimal points — exactly the geometry we predicted!

For a linear objective $f = ax + by$ on a circle, the max and min always occur at $\pm\frac{1}{\sqrt{a^2+b^2}}\langle a, b \rangle$.

Three Variables: $f(x, y, z)$ with Constraint $g(x, y, z) = k$

The method extends naturally to functions of three variables.
Optimize $f(x, y, z)$ subject to $g(x, y, z) = k$:

$$\nabla f = \lambda \nabla g \quad \Longrightarrow \quad f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad f_z = \lambda g_z$$

Plus the constraint: $g(x, y, z) = k$.
That's **4 equations in 4 unknowns** (x, y, z, λ).

Same idea, one more variable. The system is bigger but the strategy is the same: use the gradient equations to relate the variables, then plug into the constraint.

Quick Check: Set Up a 3-Variable Lagrange System

Minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + 2y + 3z = 6$.
Write the four equations you'd need to solve (but don't solve them).

Identify f and g, compute ∇f and ∇g, then write $f_x = \lambda g_x$, $f_y = \lambda g_y$, $f_z = \lambda g_z$, and $g = k$.

$\nabla f = \langle 2x,\, 2y,\, 2z \rangle, \quad \nabla g = \langle 1,\, 2,\, 3 \rangle.$
The system:

$$2x = \lambda, \quad 2y = 2\lambda, \quad 2z = 3\lambda, \quad x + 2y + 3z = 6$$

Same pattern as two variables — one more gradient equation, one more unknown. Each equation immediately gives x, y, z in terms of λ.

Example 4: Minimum Surface Area Box

Example 4

Find the dimensions of a rectangular box with volume $1000\ \mathrm{cm}^3$ that has the **smallest** surface area.

Set up:

- Surface area: $f(x, y, z) = 2xy + 2xz + 2yz \leftarrow$ minimize this
- Volume constraint: $g(x, y, z) = xyz = 1000$

Compute gradients, write the 4 equations, and look for symmetry!

Gradients:

$$\nabla f = \langle 2y + 2z,\ 2x + 2z,\ 2x + 2y \rangle$$

$$\nabla g = \langle yz,\ xz,\ xy \rangle$$

System from $\nabla f = \lambda \nabla g$:

$$2y + 2z = \lambda yz \quad (1) \qquad 2x + 2z = \lambda xz \quad (2) \qquad 2x + 2y = \lambda xy \quad (3)$$

$$xyz = 1000 \quad (4)$$

Use symmetry: Subtract (2) from (1):

$$2(y - x) = \lambda z(y - x)$$

Either $y = x$ or $\lambda z = 2$. Similarly, subtracting (3) from (2) gives $z = x$ or $\lambda y = 2$.

If $x = y = z$: From the constraint: $x^3 = 1000 \Rightarrow x = 10$.

The box is a cube: $10 \times 10 \times 10$ cm, with surface area $600\ \mathrm{cm}^2$.

The cube minimizes surface area for a given volume. Symmetry in the equations often signals that the optimal solution is symmetric too.

When to Use Lagrange Multipliers

How does Lagrange fit into the optimization toolkit from Chapter 14?

Step 1: What kind of region?

Region Type	What to Do
Open region (no boundary)	Find critical points of f; use Second Derivatives Test (14.7)
Closed & bounded region	Check **interior** critical points + optimize on the **boundary** separately

Step 2: If you need to optimize on a boundary curve $g = k$\ldots

Boundary Method	Best When
Parameterize the curve	Boundary has a simple parameterization (line, circle)
Lagrange multipliers	Constraint is hard to parameterize, or you have multiple constraints

Lagrange multipliers don't replace the critical-point method — they handle the **boundary step** that the critical-point method can't.

Homework

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